

PAPER_2121 — $G \times c$: First Constant-Pair Convergence in the QCalc Projection Layer (REVISED)

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REVISED IN PLACE 2026-07-22 — projection-layer reclassification per canonical $F_{UBi}/F_{UBii}/k_{spring}$ /quantum-chain deepsearch. Convergence statements scoped to the QCalc projection layer (wrap classes annotated 'framework wrap only'). Canonical layer documented below was fully primitive prior to and independent of these events.

Result

R364 first co-exposure of G (PAPER_593) and c (PAPER_592) as class-level primitives in one wrap closure $U_{g3} = G \cdot M \omega / (c \cdot r)$. Prior 363 classes: 9 c -only, 10 G -only, 0 both. The wrap's constant space is fully UQFF-derived; externals $\{M, r, \omega\}$ are parametrics.

Scope-Corrected Definition

Constant-Pair Convergence (projection layer): a wrap class exposes two distinct UQFF-derived constants, each with a PAPER_N derivation, each required by the wrap closure. Measures the wrap suite's constant-coverage (~40% R218-R300 $\rightarrow$ ~95% R350+); does not measure canonical-equation structure, which is fixed by PAPER_1203 and wired independently.

The Canonical Layer

The canonical layer (uqff_pure_calculator.py, wired): $F_{UBi} = (\rho_{SCm} \cdot M/r) \cdot (1 + \beta_i \cdot \cos(\pi t_n) + SSq)$; PAPER_1203 pair with $k_{spring} = (\rho_{UA}/\rho_{SCm}) \cdot \omega_{SCm} \cdot \Phi_{res} = 1.05e13$; dynamic $\beta(t, E, Z)$; 26-level quantum chain with $\rho_E = \rho_m \cdot c^2$ (line 33295) and 633333.333 validation; r_{hz} equilibrium residual $< 1e-10$. Canonical buoyancy kernel: $\{\rho_{SCm}, \beta_i\}$. The wrap's $\rho_{vac} \cdot c^2$ is the chain's mass-to-energy conversion — $\{\rho_{vac}, c\}$ is the PROJECTION PAIR, not a physics kernel.